Installation Red Hat Enterprise Linux

You must have an active Red Hat subscription to download **RHEL 7.0 ISO** image from Red Hat Customer Portal.

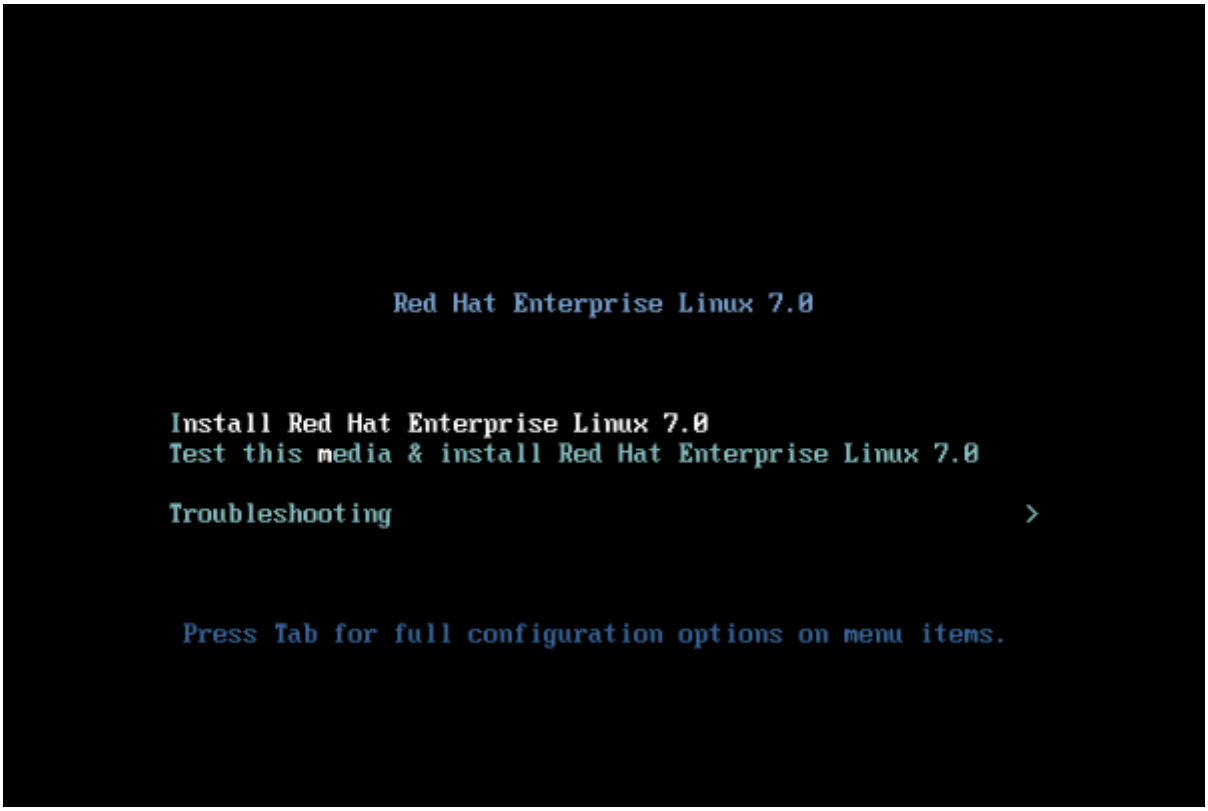
1. RHEL 7.0 Binary DVD ISO image

Although RHEL can be installed on a variety of platforms, such as AMD 64, Intel 64, IBM System Z, IBM Power, etc. This tutorial covers the **RHEL 7.0** basic minimal installation with on an Intel **x86-64** processor architecture using a binary **DVD ISO** image, an installation best suited for developing a high customizable server platform with no Graphical Interface.

Installation of Red Hat Enterprise Linux 7.0

1. After registering on **Red Hat Customer Portal** go to Download section and grab the last version of **RHEL DVD Binary ISO** image, then burn it to a DVD media or create a USB bootable media using Unetbootin LiveUSB Creator.

2. Then place the DVD/USB in your appropriate system drive, start your computer, select bootable unit and on the first RHEL prompt select **Install Red Hat Enterprise Linux 7.0**.

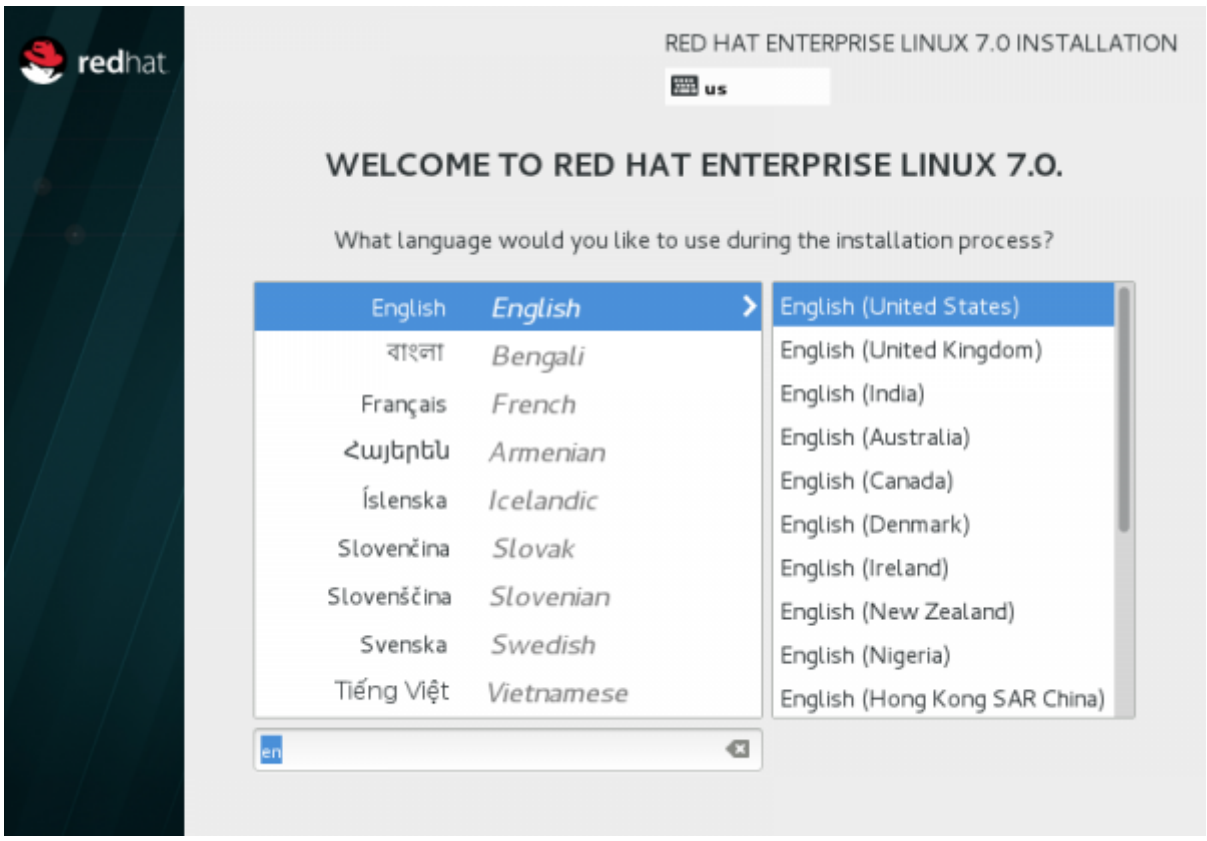


Install Red Hat Enterprise Linux 7.0

3. After the system loads, select the **language for installation** process and hit on **Continue**.

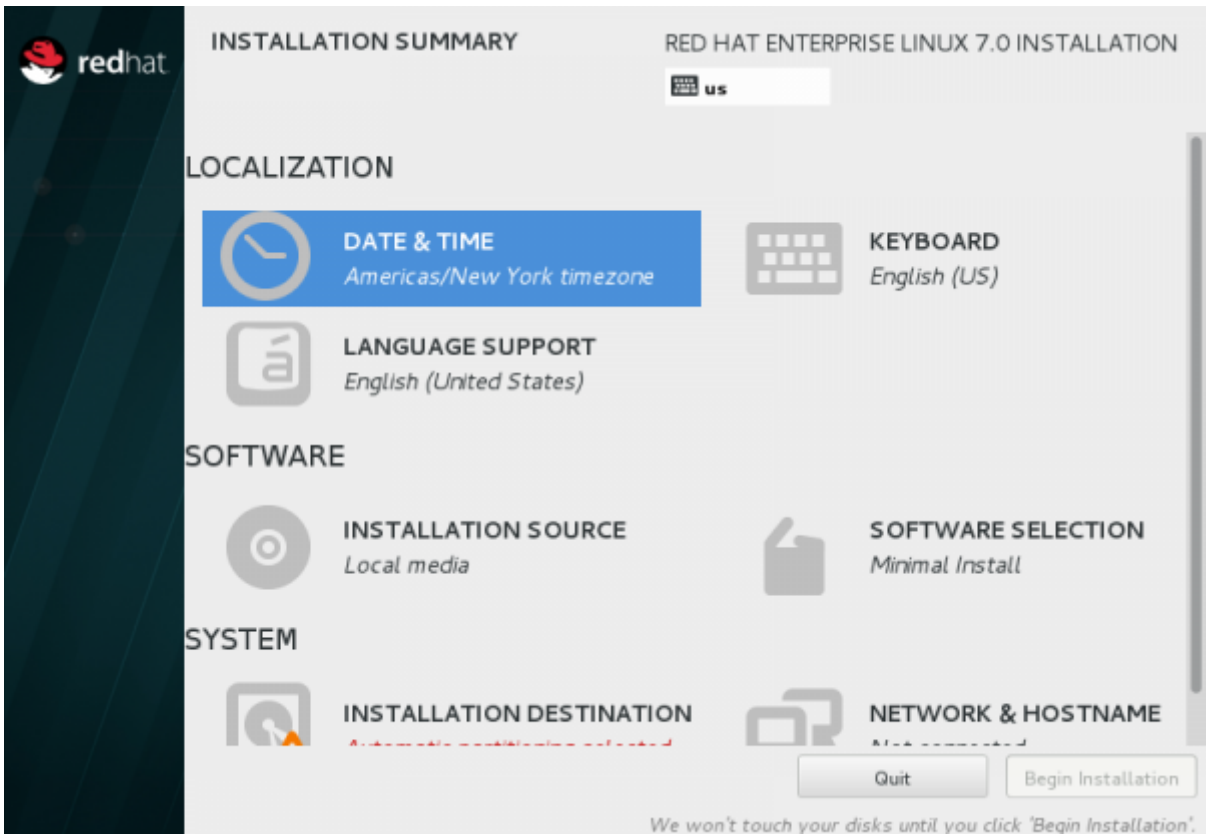
```
[  OK  ] Created slice system-anaconda\x2dtmux.slice.
[  OK  ] Created slice system-anaconda\x2dshell.slice.
[  OK  ] Listening on LVM2 metadata daemon socket.
         Starting Apply Kernel Variables...
         Mounting Debug File System...
         Mounting POSIX Message Queue File System...
         Mounting Huge Pages File System...
[  OK  ] Set up automount Arbitrary Executable File Formats File System Automount Point.
         Starting Create list of required static device nodes for the current kernel...
[  OK  ] Reached target Paths.
         Starting Device-Mapper Multipath Device Controller...
[  OK  ] Listening on udev Kernel Socket.
[  OK  ] Listening on udev Control Socket.
         Starting udev Coldplug all Devices...
[  OK  ] Reached target Swap.
         Starting Remount Root and Kernel File Systems...
         Mounting Temporary Directory...
[  OK  ] Started Remount Root and Kernel File Systems.
         Starting Import network configuration from initramfs...
         Starting Configure read-only root support...
         Starting Load/Save Random Seed...
[  OK  ] Stopped Trigger Flushing of Journal to Persistent Storage.
         Stopping Journal Service...
[  OK  ] Stopped Journal Service.
         Starting Journal Service...
[  OK  ] Started Journal Service.
[  OK  ] Started Apply Kernel Variables.
[  OK  ] Started Load/Save Random Seed.
[  OK  ] Started Create list of required static device nodes for the current kernel.
         Starting Create static device nodes in /dev...
[  OK  ] Mounted Temporary Directory.
[  OK  ] Mounted Debug File System.
[  OK  ] Mounted POSIX Message Queue File System.
[  OK  ] Mounted Huge Pages File System.
[  OK  ] Started udev Coldplug all Devices.
```

Booting RHEL 7

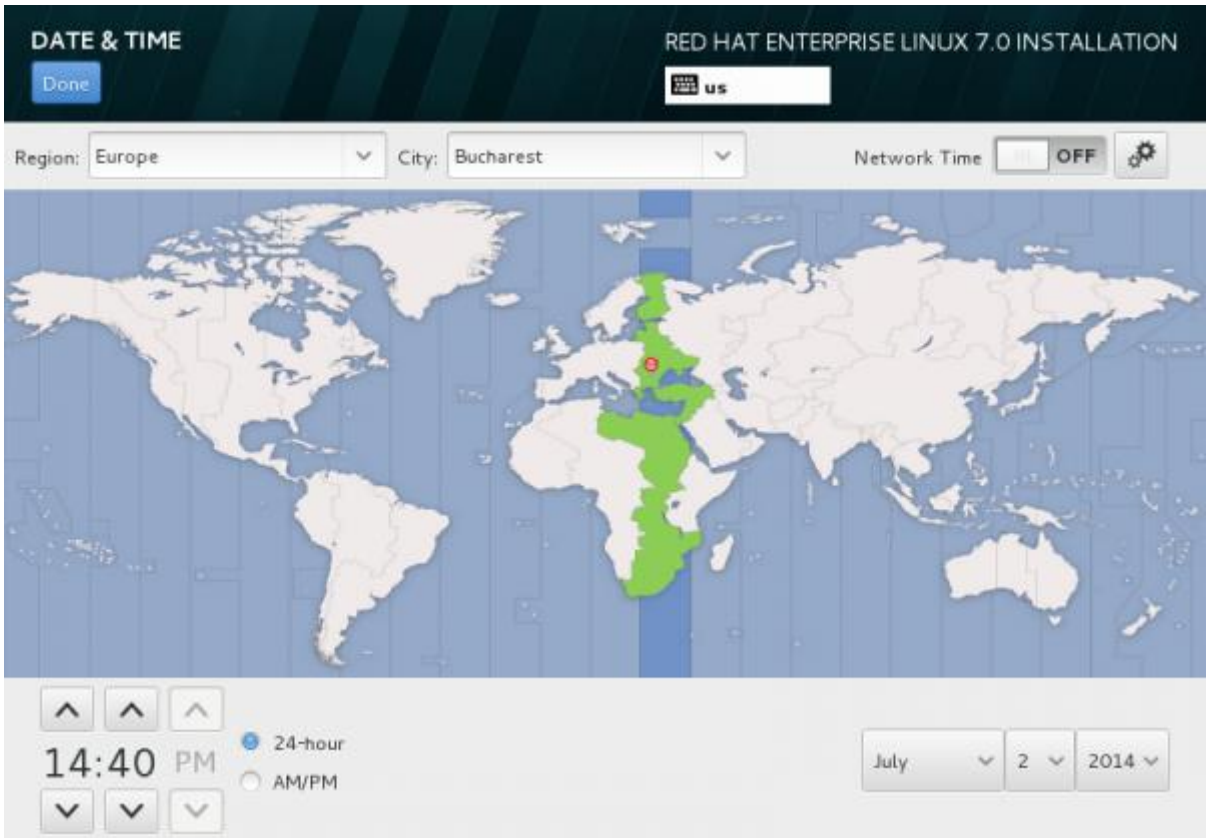


Select Language for Installation

4. When the installer gets on **Installation Summary** it's time to customize the installation process. First click on **Date & Time**, choose your system location from the provided map and hit on **Done** to apply configuration.

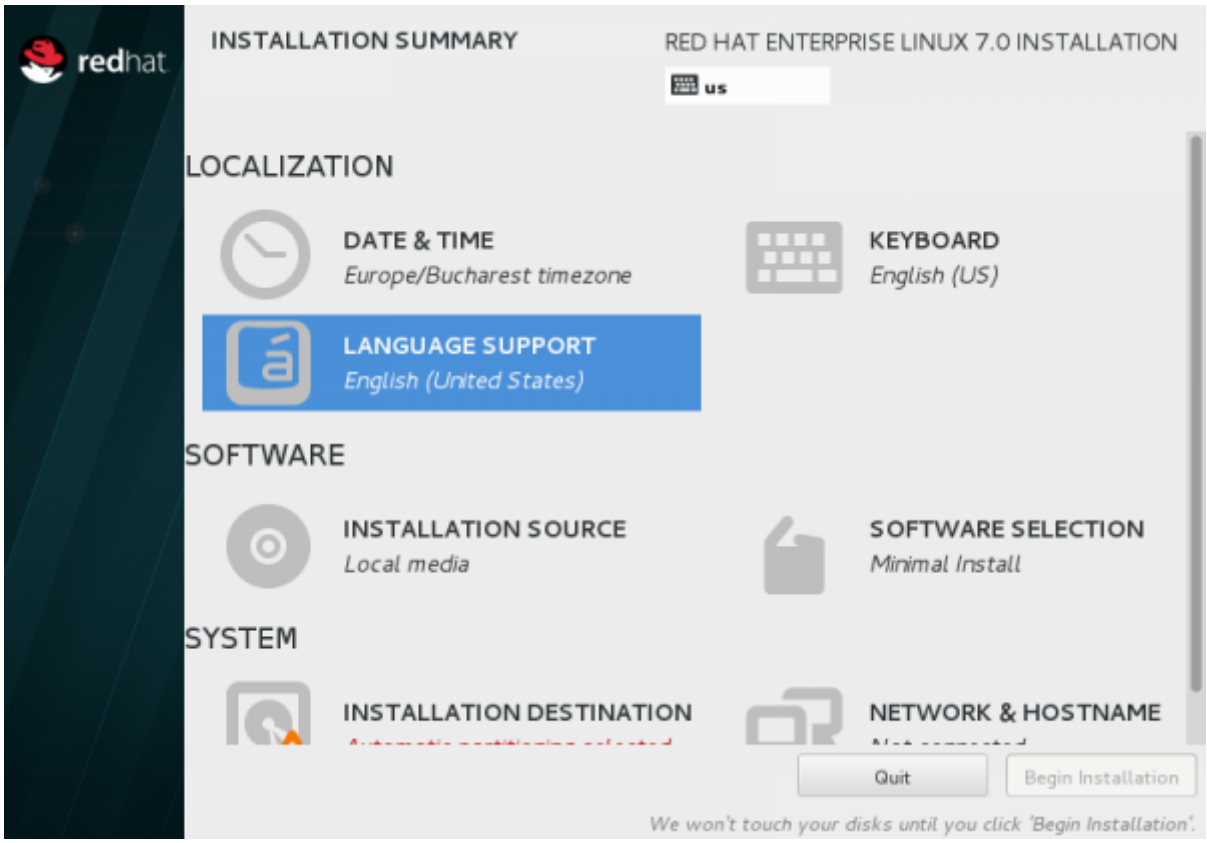


RHEL 7 Installation Summary



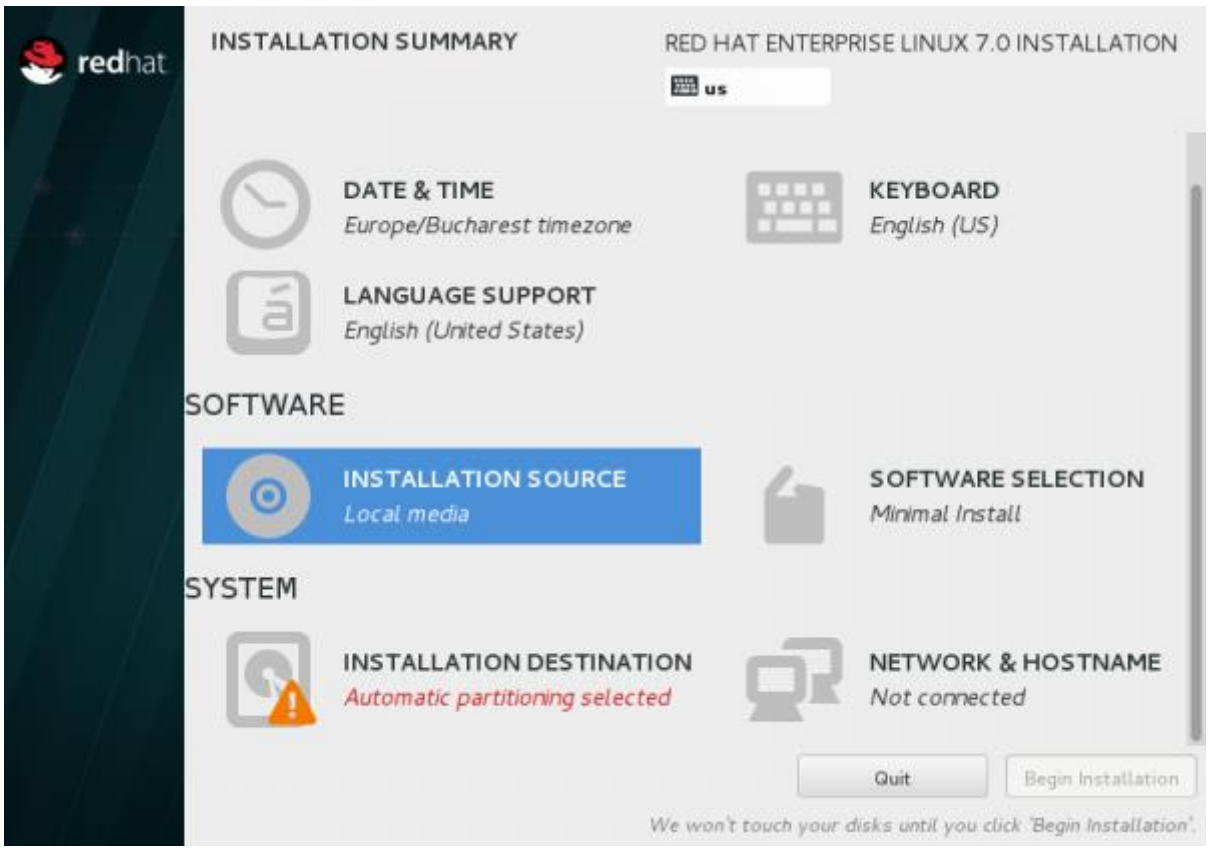
Select Date, Time and Location

5. The next step is to change **Language System Support** and **Keyboard** language. Click on both if you want to change or add other languages to your system but for a server the recommendation is to stick with English language.

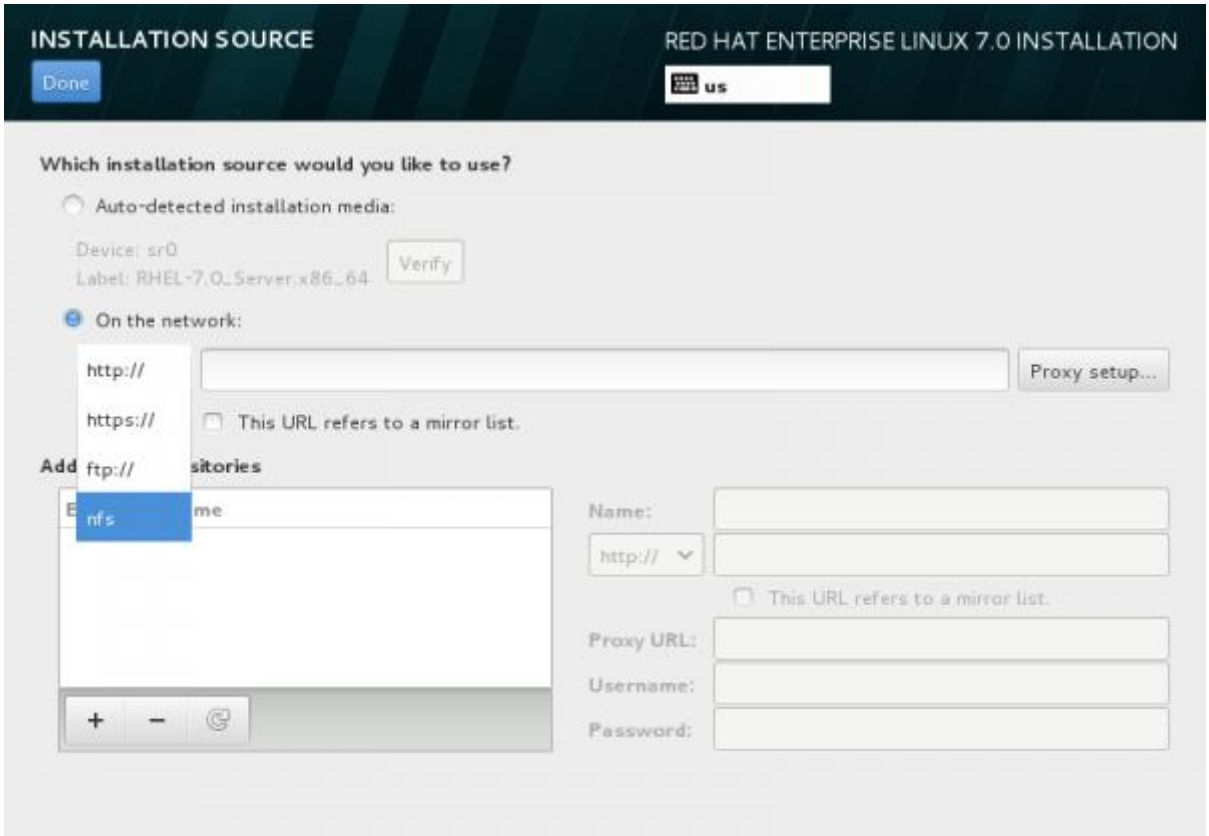


Language System Support

6. If you want to use other sources than the ones provided by the DVD media hit on **Installation Source** and add your **Additional Repositories** or specify a network location using **HTTP**, **HTTPS**, **FTP** or **NFS** protocols then hit on **Done** to use your new sources. If you can't provide other sources stick to default one **Auto-detected installation media**.

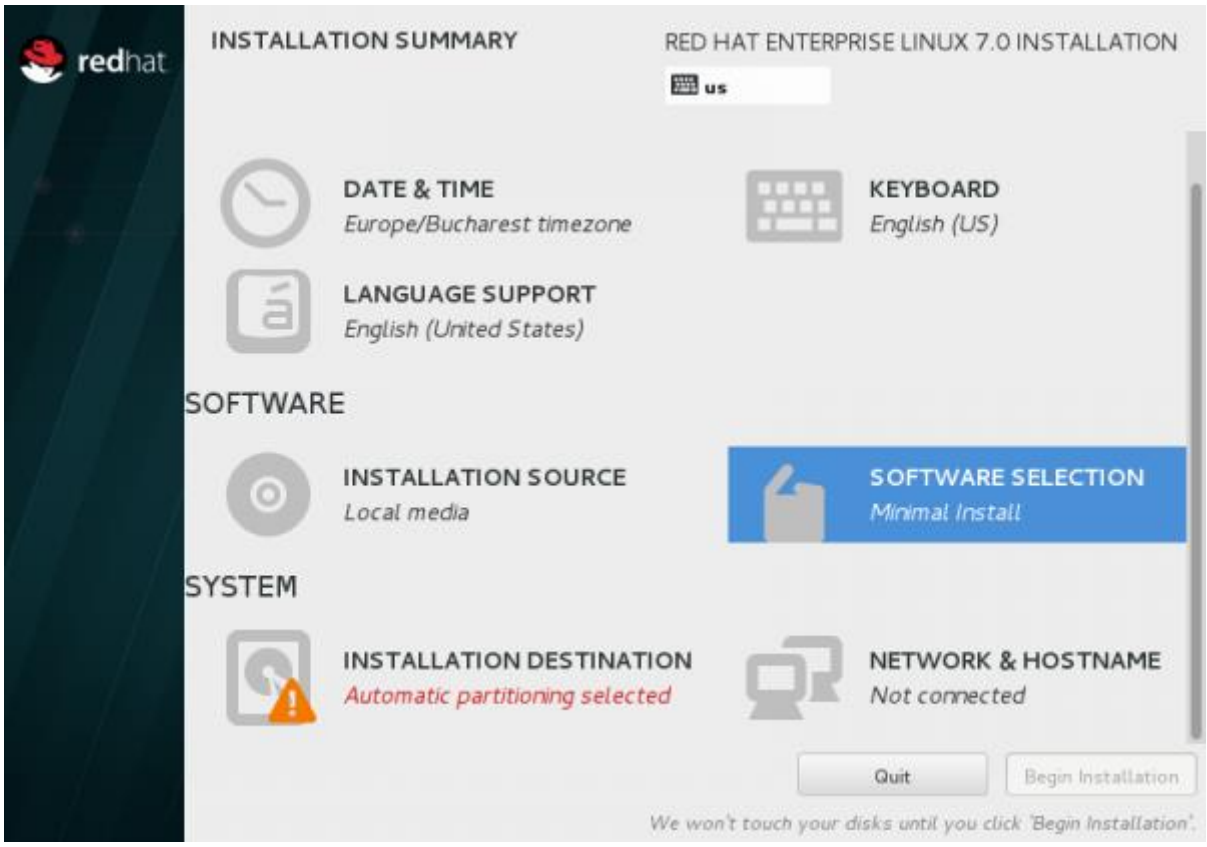


Installation Source

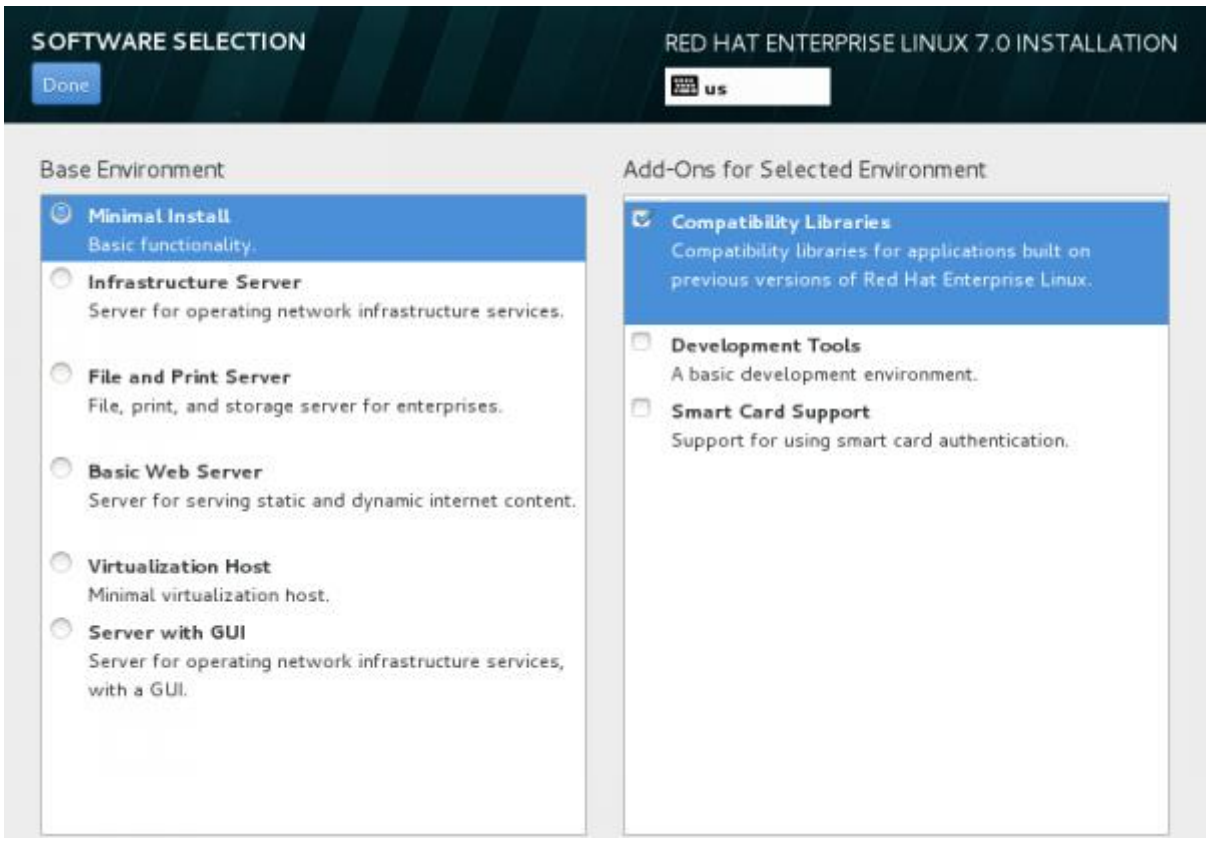


Select Network Location

7. The next important step is to select your system software. Click on **Software Selection** and choose your Base Installation Environment from the down-list. For a highly customizable platform where you can install only the packages that you need after the installation, choose **Minimal Install** with **Compatibility Libraries Add-ons**, then hit on **Done** to apply this changes to installation process.



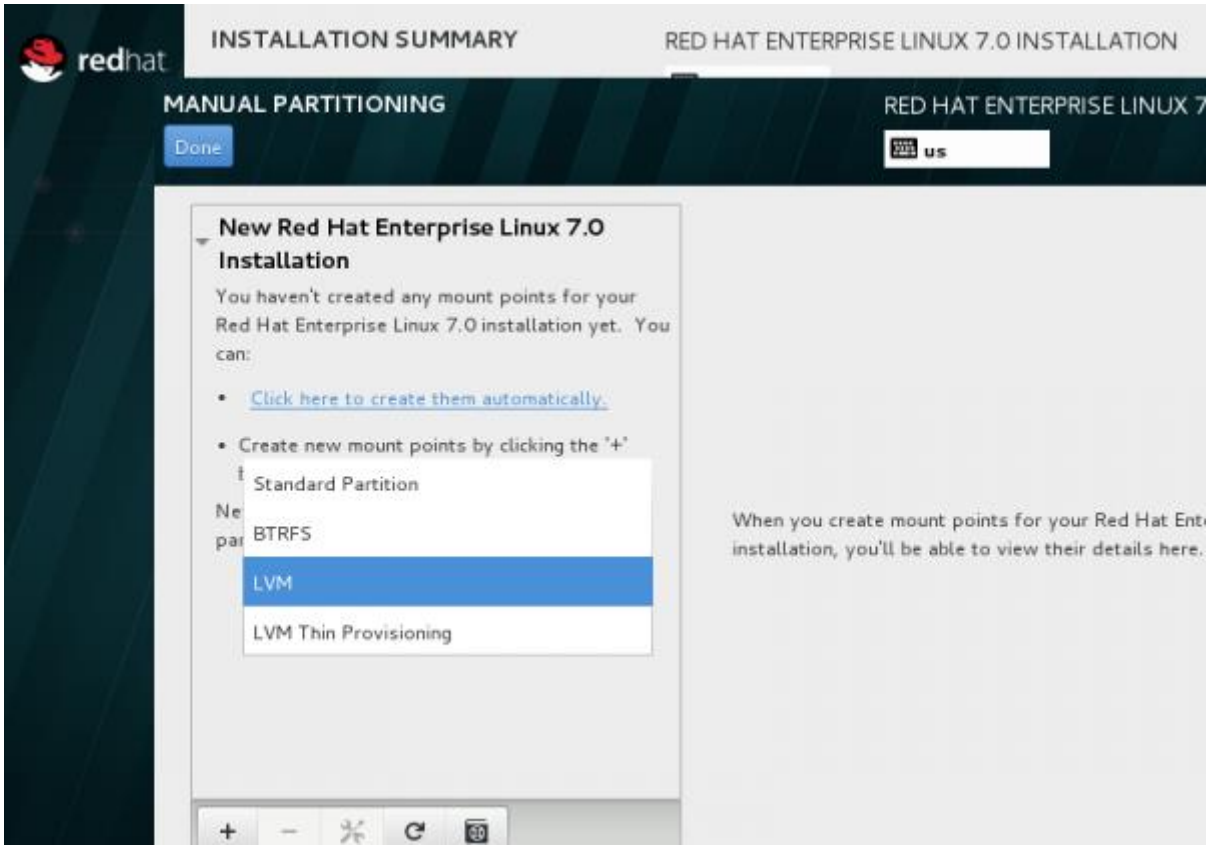
Select Software Selection



Select RHEL 7 Minimal Installation

8. The next important step is to configure your system partitions. Click on **Installation Destination**, select **LVM** as partition scheme for a
better management over system space, then hit on **Click here to create them automatically**.

Installation Destination



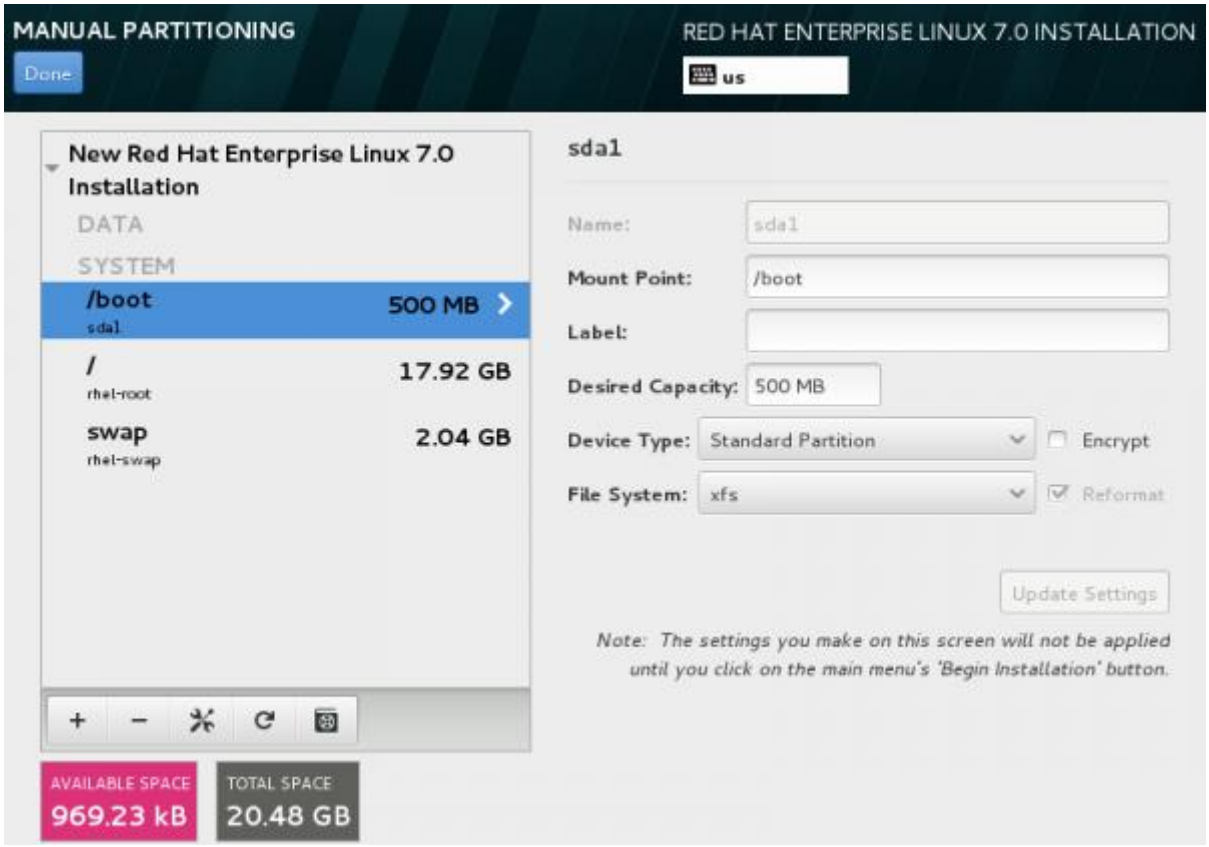
Select LVM Partition Scheme

9. After the installer presents you with default system partition scheme you can edit in any way that suits you (delete and recreate partitions and mount points, change partitions space capacity and file system type, etc.). As the base scheme for a server you should use dedicated partitions such as:

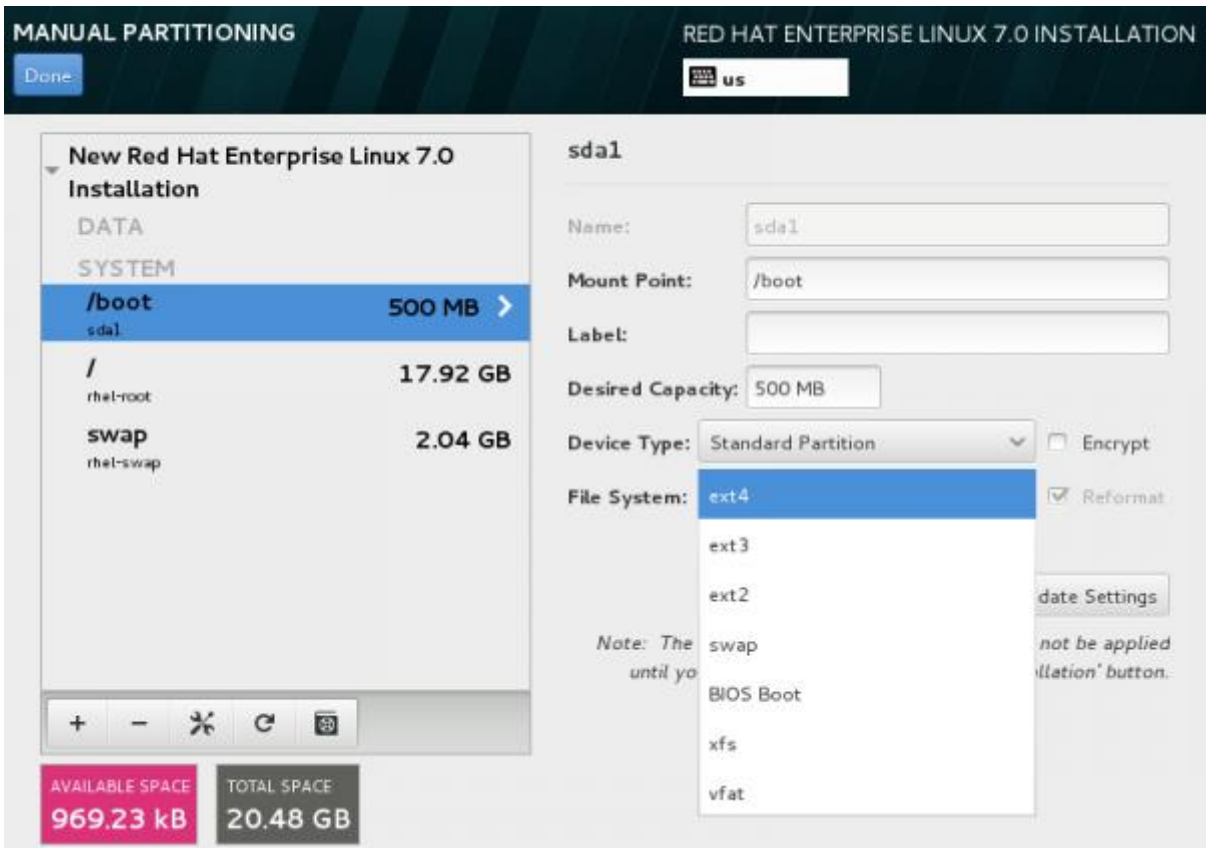
1. /boot – 500 MB – non-LVM
2. /root – min 20 GB – LVM
3. /home – LVM
4. /var – min 20 GB – LVM

With **XFS** filesystem, which is the most advanced filesystem in the world. After editing partitions hit on **Update Setting** button, then click on **Done** then **Accept Changes** on **Summary of Changes** prompt to apply new configurations.

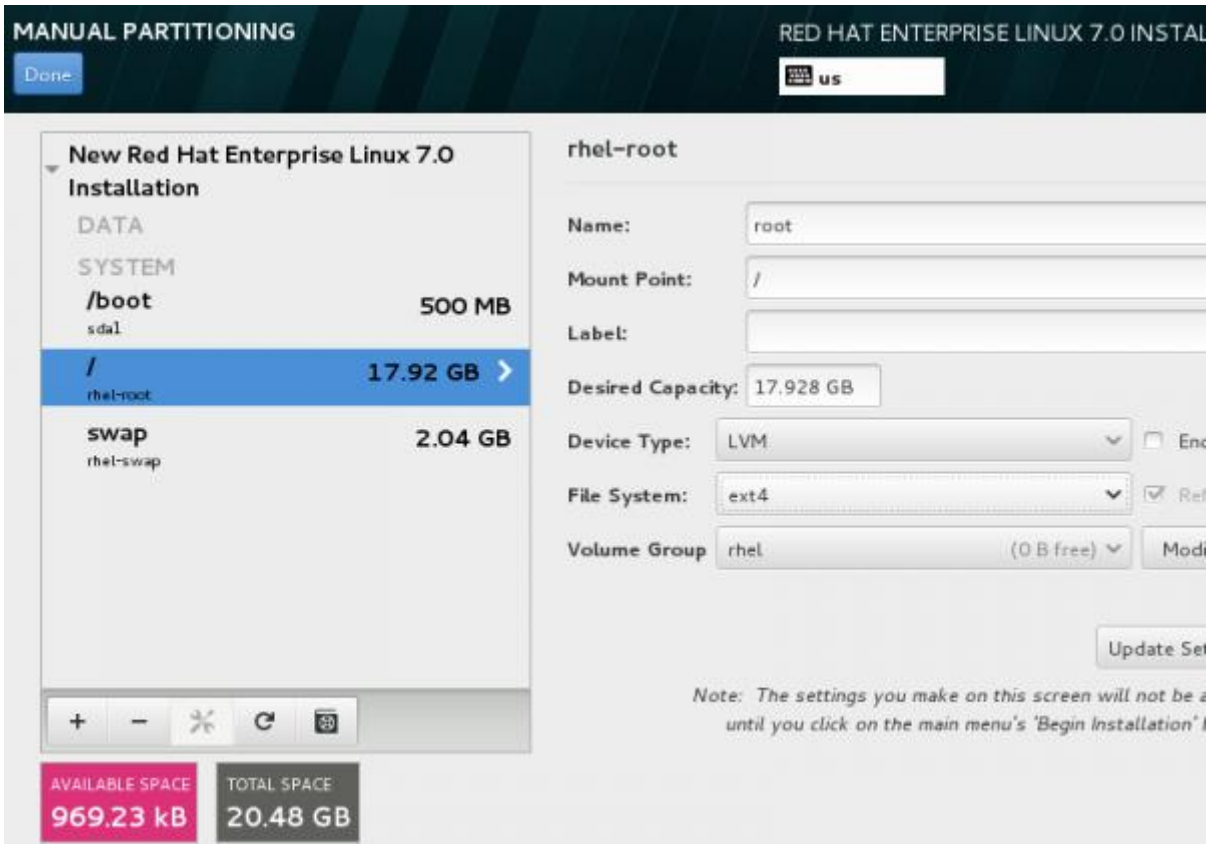
As a note, if your Hard-Disk is larger than 2TB in size the installer automatically will convert partition table to **GPT** disks and if you want to use GPT table on disks smaller than 2TB, then you should pass the argument **inst.gpt** to the boot command line in order to change the default behaviour.



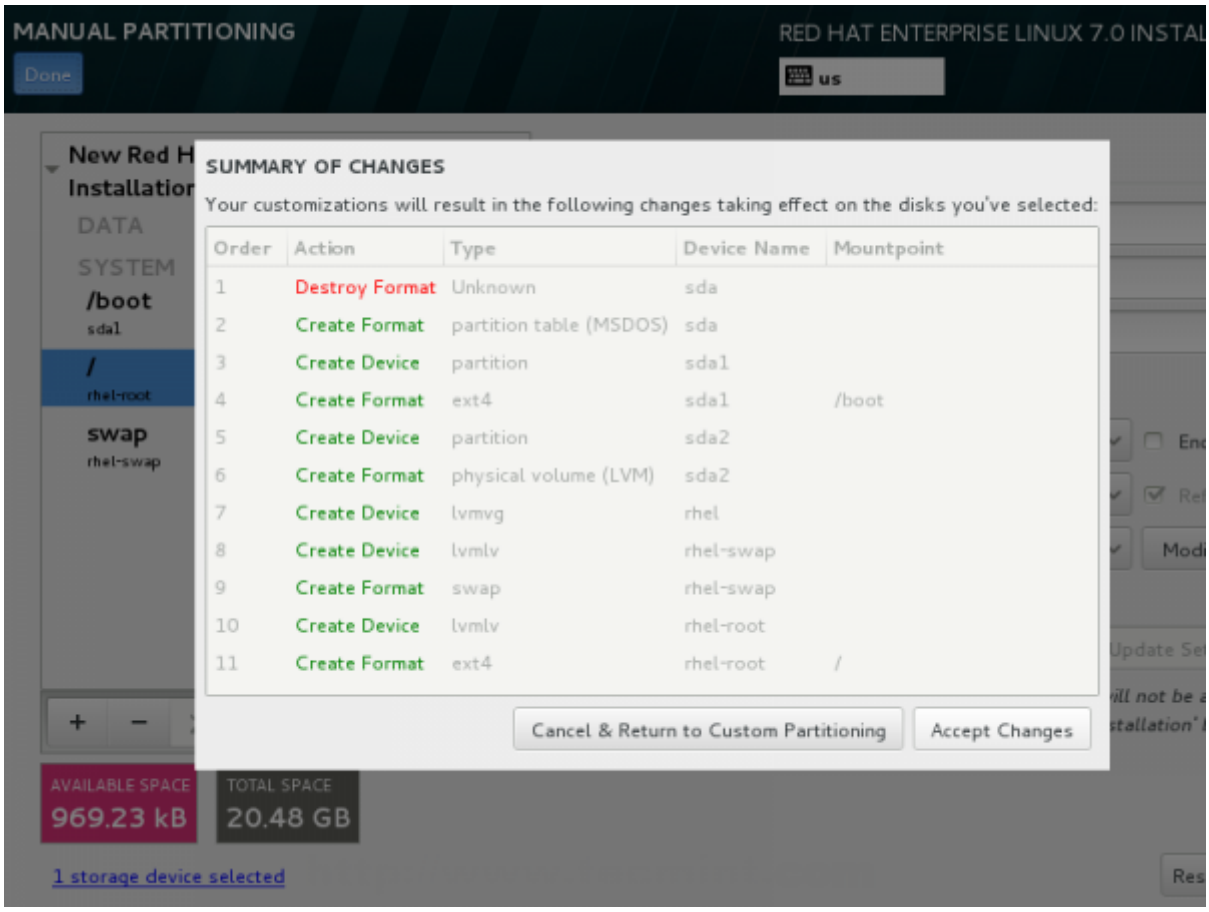
Create /boot Partition



Select /boot Partition Type

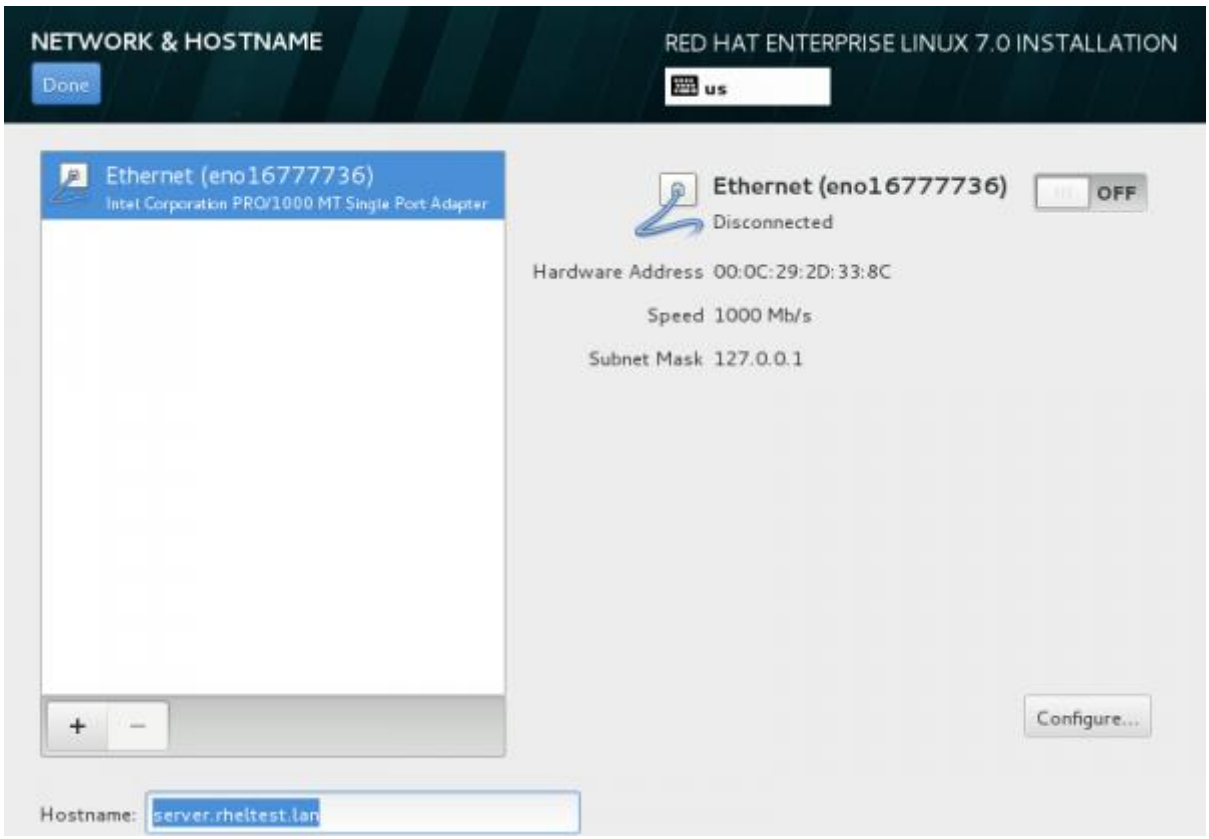


Create /root Partition



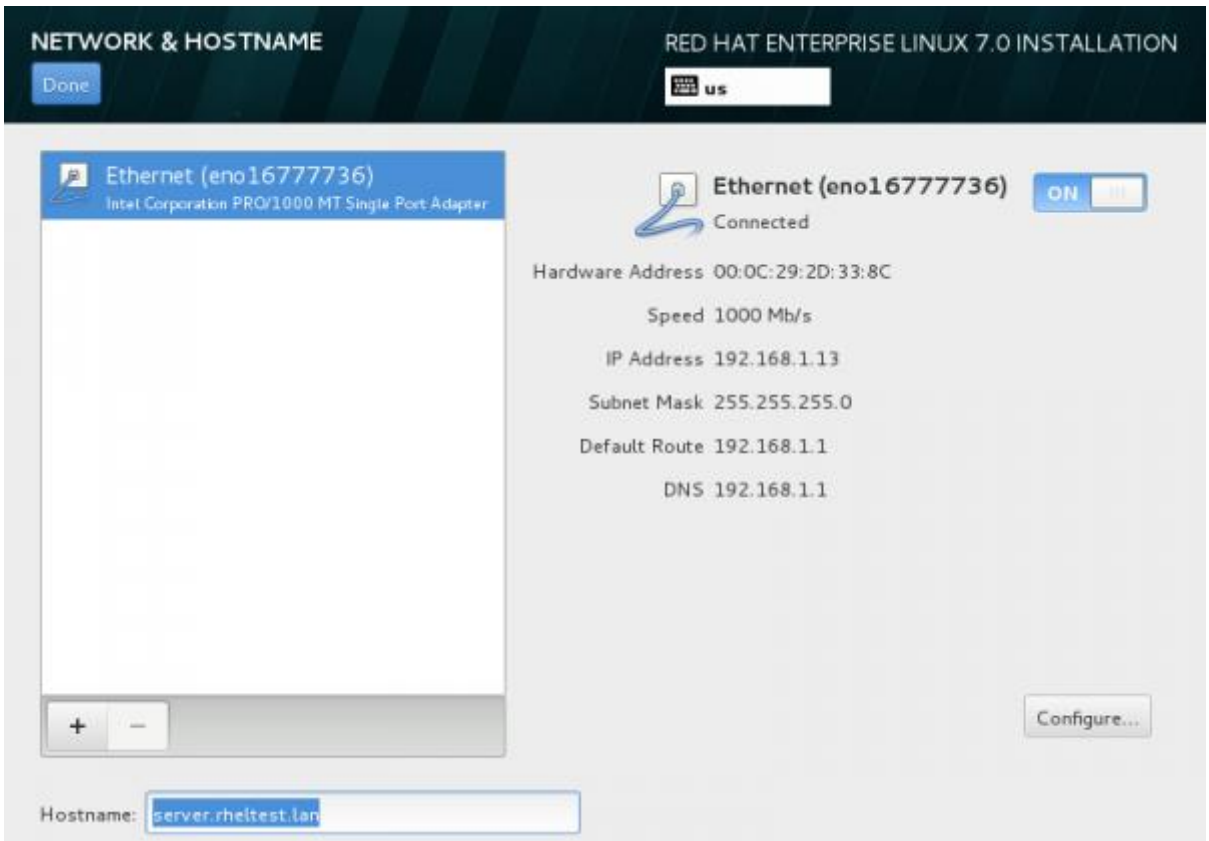
Summary of Partition Changes

10. The last step before continuing with installation process is setting your Network Connection. Click on **Network & Hostname** and setup your system hostname. Here you can use your short system hostname or you can append the dot domain (FQDN).

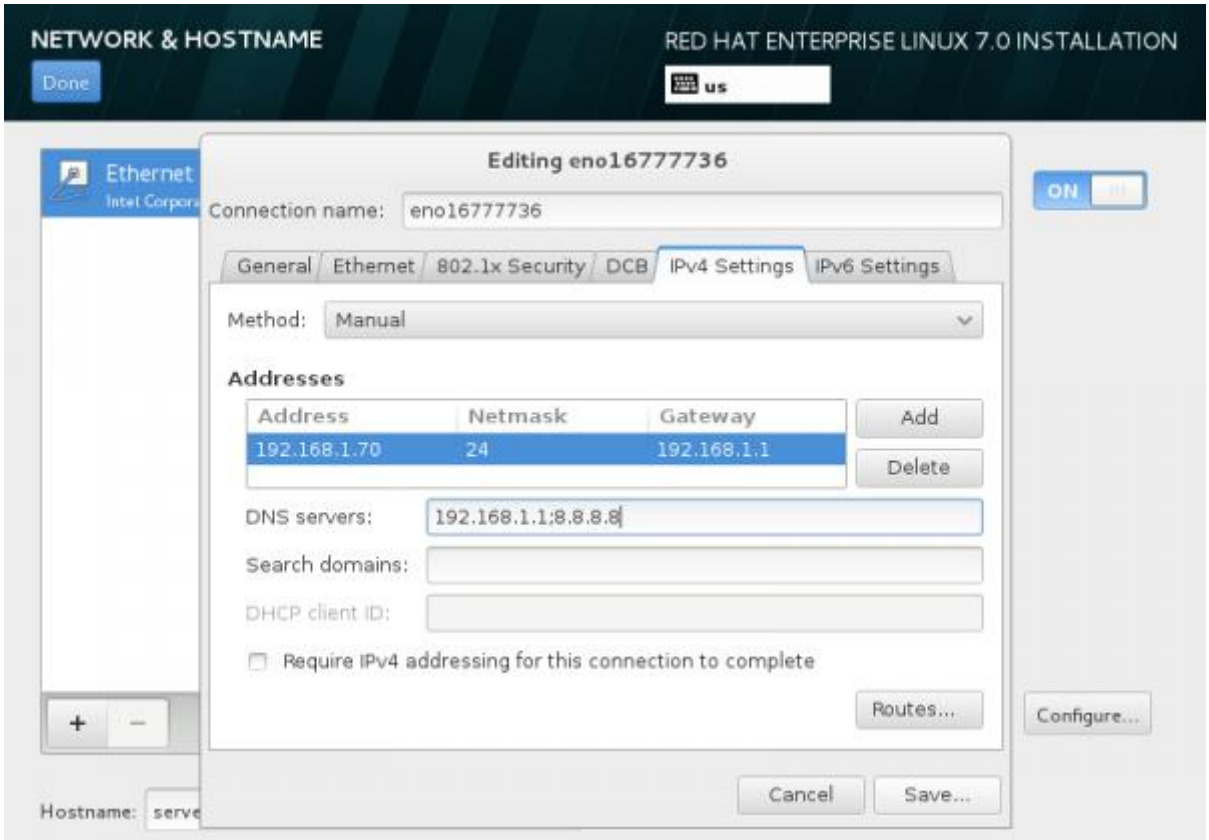


Set System Hostname

11. After setting up hostname bring up your Network Interface by switching the top **Ethernet** button to **ON**. If your network provides automatic Interface configurations through a DHCP server your IPs should be visible on Ethernet Interface Card else go to **Configure** button and provide your static network settings for your appropriate network connection.

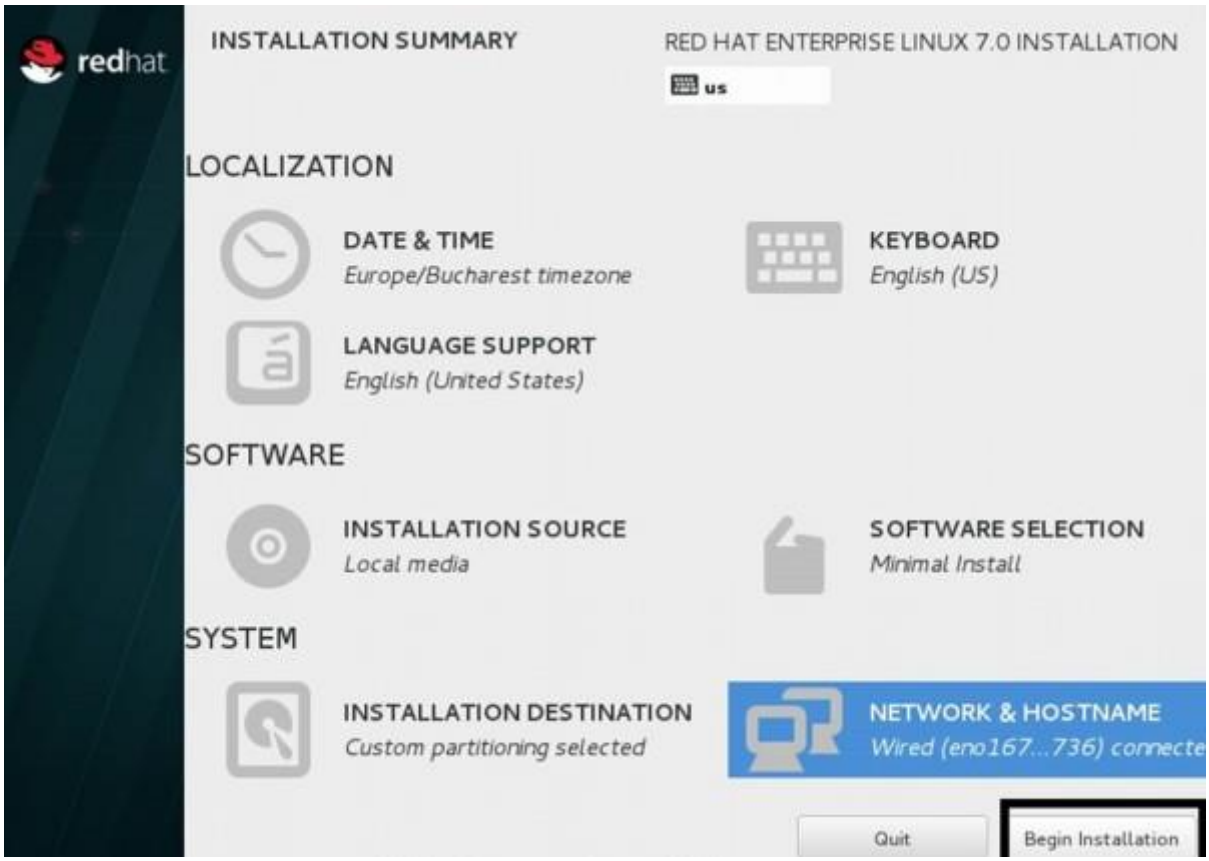


Enable Ethernet Interface



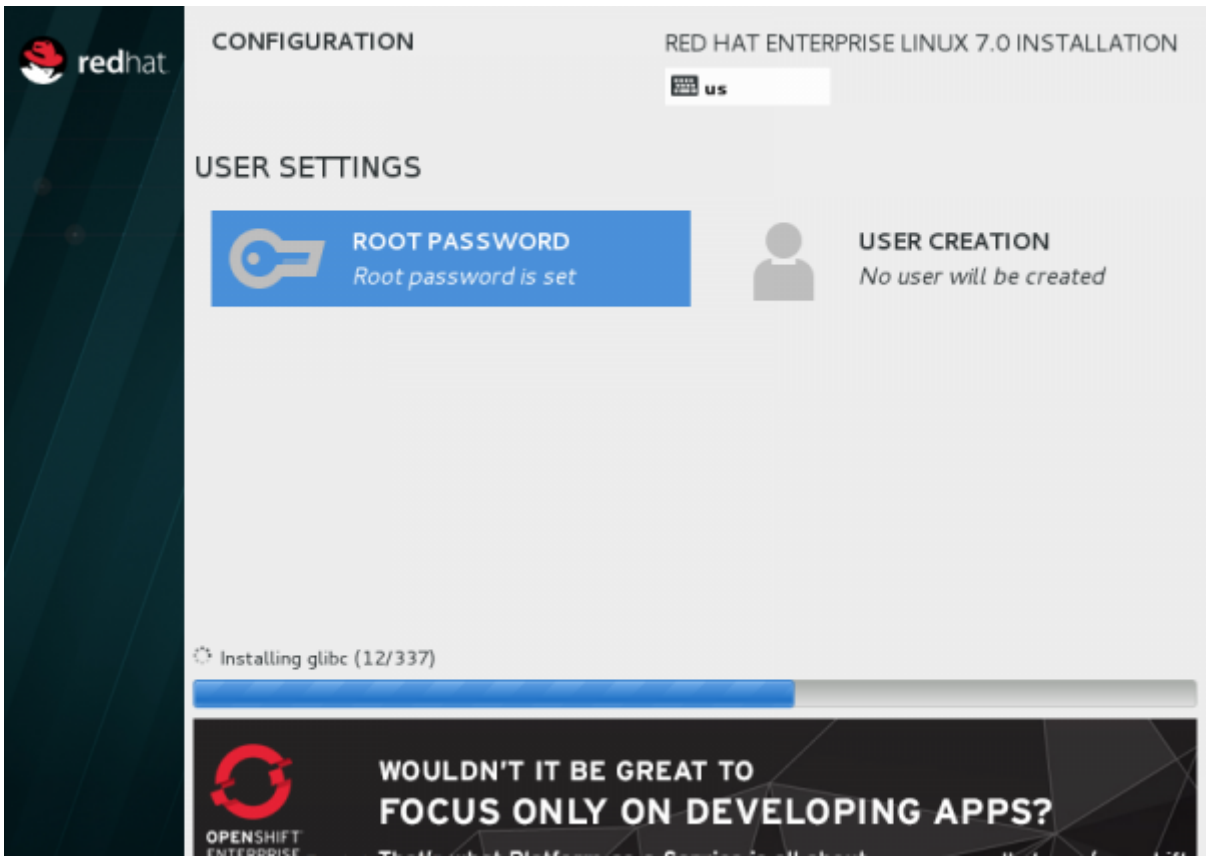
Enter Static Network Settings

12. After finishing editing Ethernet Interface settings hit on **Done** which you bring you to default window installer and after you check on your installation settings hit on **Begin Installation** to proceed further with system installation.

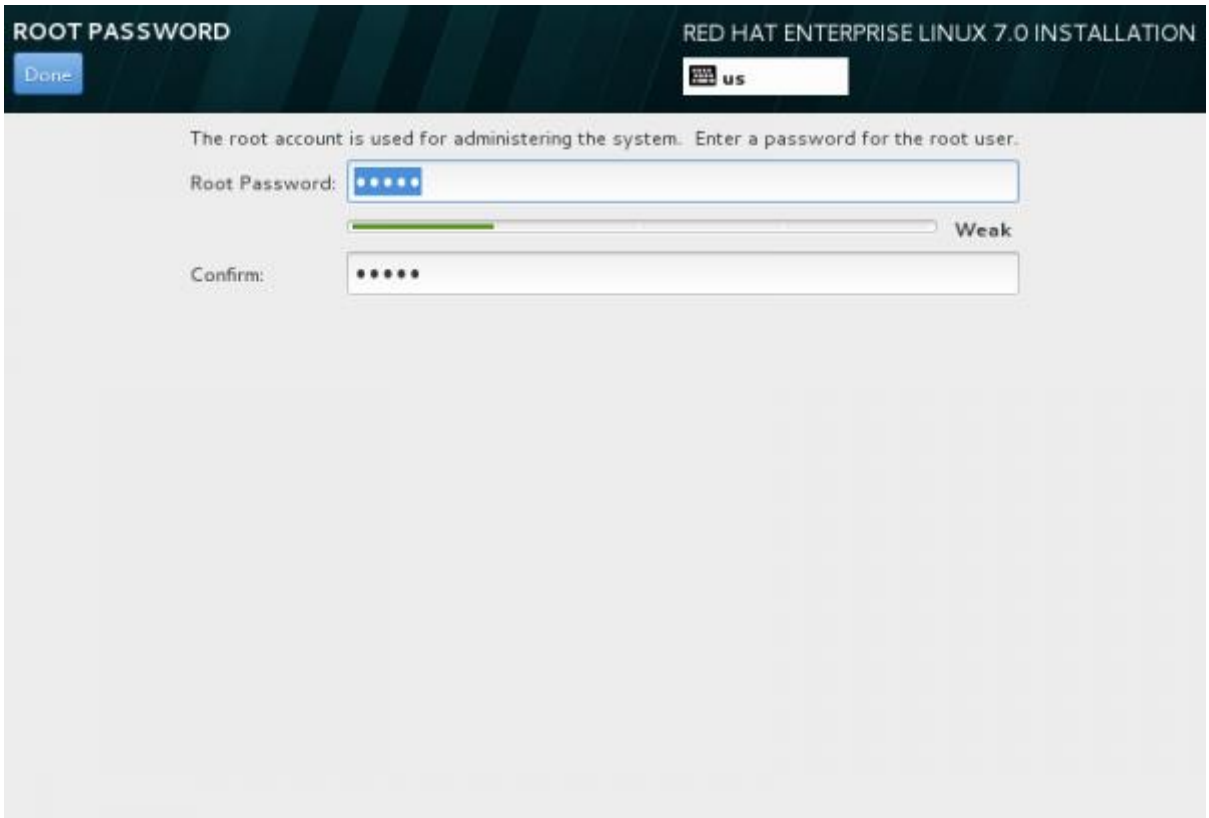


Begin RHEL 7 Installation

13. As the installation begins writing the system components on your hard-disk, you need to supply your **Root Password** and create a new User. Click on **Root Password** and try choose a strong with one at least eight characters in length (alpha-numerical and special characters) and hit in **Done** when you finish.

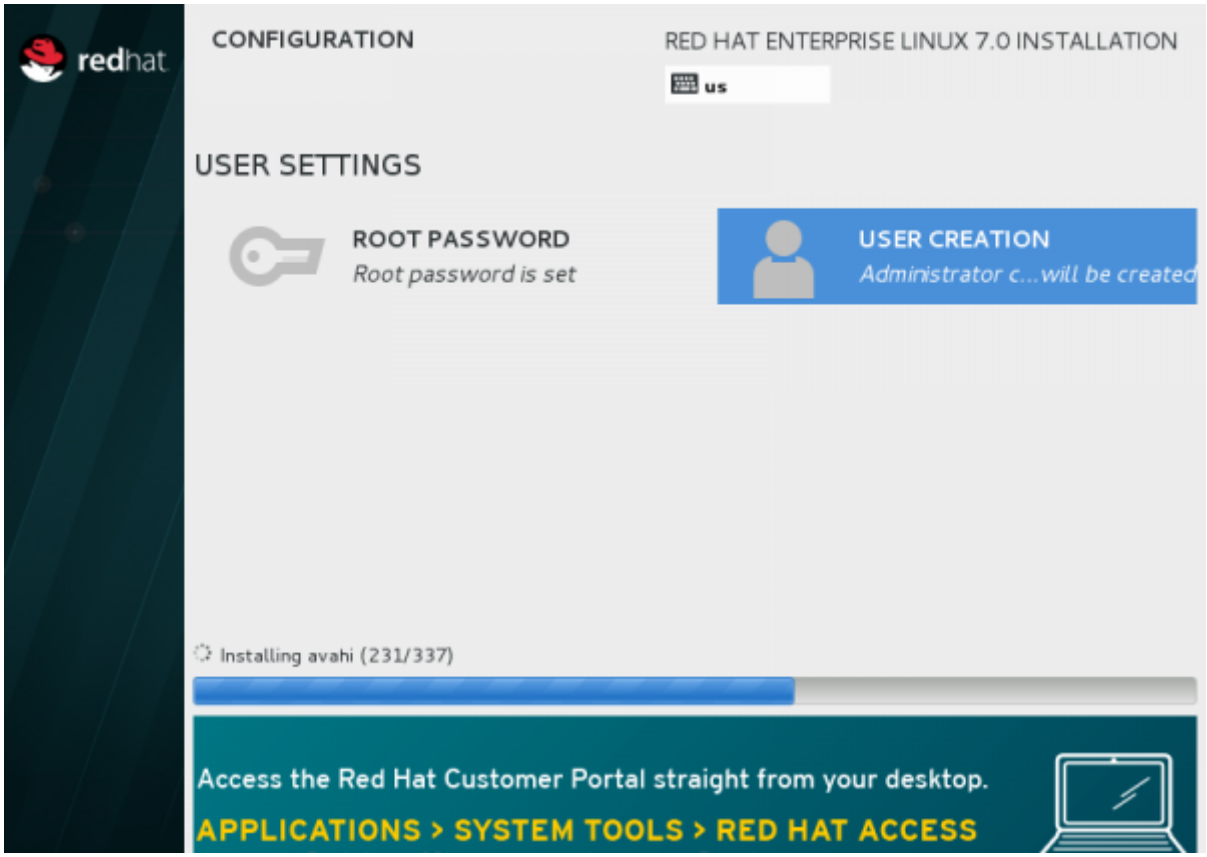


RHEL 7 Installation Process

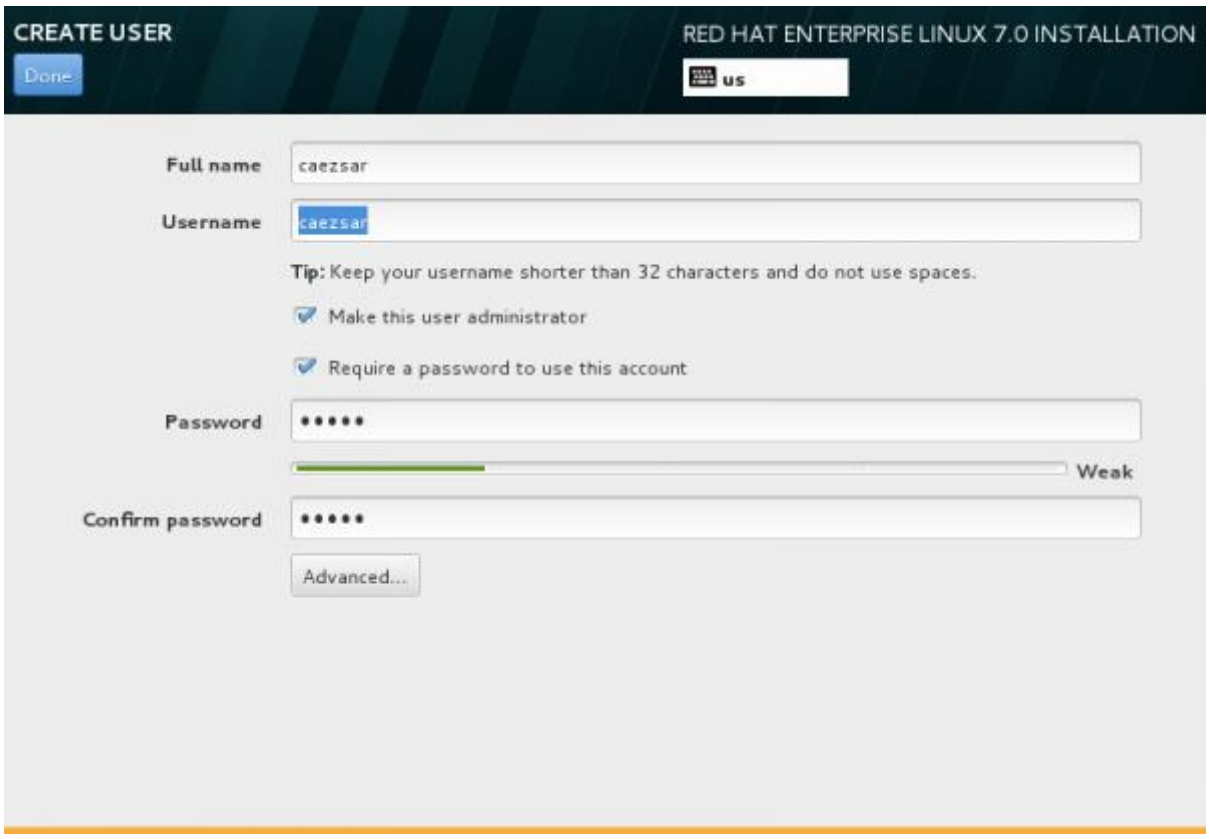


Set root Password

14. Then move to **User Creation** and provide your credentials for this new user. A good idea is to use this user as a system administrator with root powers through **sudo** command by checking the box **Make this user administrator**, then click on **Done** and wait for the installation process to finish.

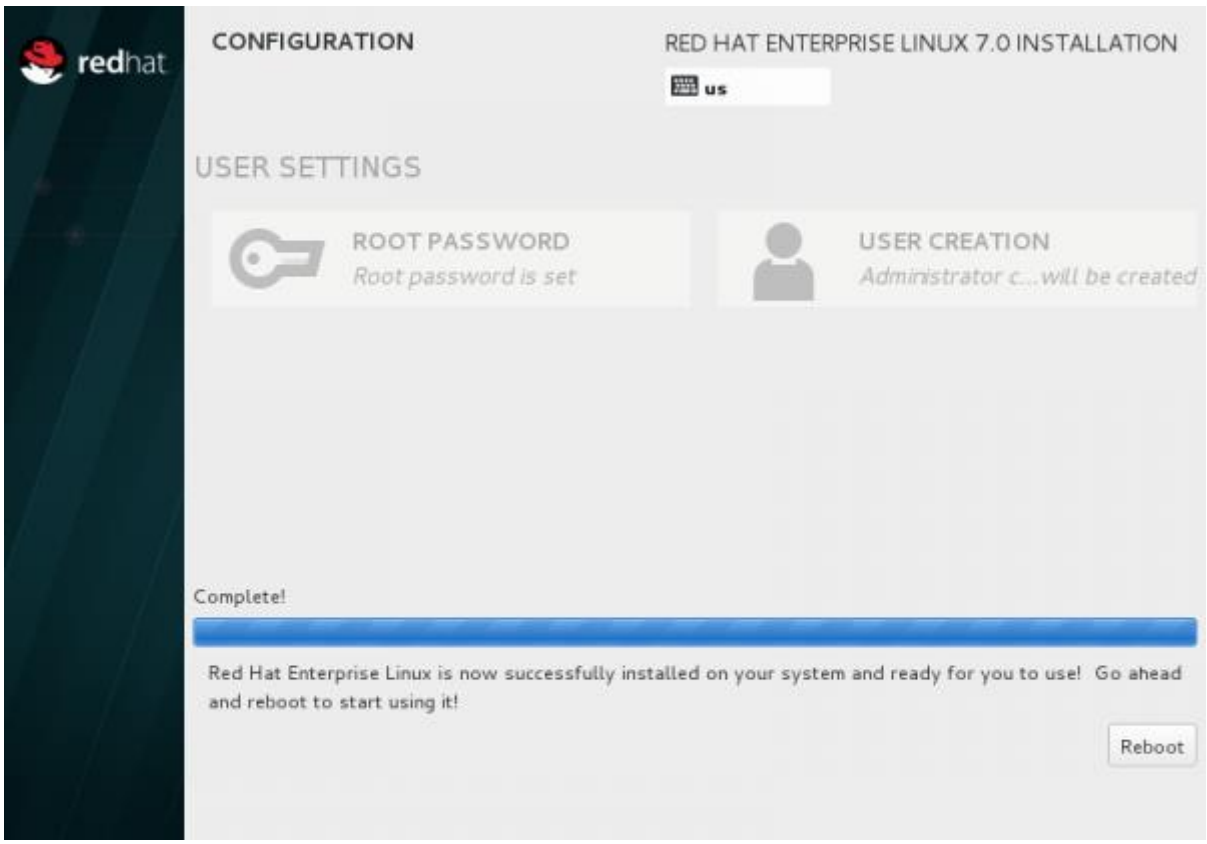


Select User Creation



Enter User Details

15. After the installation finishes the installer will announce that everything completed with success so you should be ready to use your system after reboot.



RHEL 7 Installation Complete